

SIMATS ENGINEERING
CO2_ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1522

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COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)*

Dave's Taxonomy: Manipulation (Level 2)

Total Marks: 50

Question Count: 5

Code of Conduct

I M . Kasif Jalal (192424214) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____M.kasif jalal_____

Assessment Tasks

Analytical Problem Solving

1. Compare Type-1 and Type-2 hypervisors for a cloud datacenter hosting mission-critical applications. Analyze performance, security, and manageability, then justify the better choice.
2. A cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same hardware. Analyze how virtualization architecture supports isolation and identify two major attack surfaces.
3. Study the following VM host utilization snapshot and recommend corrective actions: CPU 92%, memory 88%, storage I/O latency 35 ms, average VM boot time 170 s. Explain the likely bottlenecks.
4. Analyze how Software Defined Datacenter architecture improves agility compared with a traditional datacenter. Support your answer with compute, storage, and network virtualization considerations.
5. A multi-cloud federation project fails due to identity mismatches between providers. Analyze the issue and propose a secure identity and privacy design using federation concepts.

Analytical Problem Solving Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 – Proficient	Level 2 - Developing	Level 1 - Beginning
Problem Interpretation	20%	Interprets all technical requirements accurately	Interprets most requirements correctly	Partial interpretation	Misinterprets the problem
Analytical Reasoning	25%	Strong reasoning with clear cause-effect analysis	Good reasoning with minor gaps	Basic analysis only	Weak or no analysis
Method Selection	20%	Chooses highly appropriate virtualization and security concepts	Chooses mostly suitable concepts	Partially suitable concepts	Inappropriate approach
Solution Quality	20%	Provides feasible and technically sound solution	Reasonable solution with minor issues	Basic solution with limitations	Unclear or invalid solution
Presentation of Analysis	15%	Well-structured and technically precise	Generally clear	Somewhat unclear	Difficult to follow

Total Marks Awarded:

1. Compare Type-1 and Type-2 hypervisors for a cloud datacenter hosting mission-critical applications. Analyze performance, security, and manageability, then justify the better choice.

Type-1 Hypervisor :

- Type-1 hypervisor is installed directly on the physical hardware, so it does not need a host operating system.
- Since it works directly with the hardware, it delivers faster performance and uses system resources efficiently.
- It provides strong security by keeping virtual machines isolated from each other.
- It is highly reliable and can handle large workloads without affecting performance.
- It is easy to manage multiple virtual machines in a cloud datacenter.
- Therefore, it is widely used for enterprise and cloud computing environments.
- Examples: VMware ESXi, Microsoft Hyper-V, Xen.

Type-2 Hypervisor :

- Type-2 hypervisor is installed on top of an existing operating system, such as Windows or Linux.
- Because it depends on the host operating system, it has slightly lower performance than Type-1.
- Its security also depends on the security of the host operating system.
- It is simple to install and is mainly used for learning, software testing, and application development.
- It is suitable for personal computers and small-scale environments rather than cloud datacenters.
- Examples: VirtualBox, VMware Workstation.

Justification :

- For a cloud datacenter hosting mission-critical applications, Type-1 hypervisor is the best choice.
- It offers better performance, stronger security, efficient resource management, and higher reliability, making it ideal for running critical business applications continuously.

2. A cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same hardware. Analyze how virtualization architecture supports isolation and identify two major attack surfaces.

Virtualization Architecture :

- Virtualization allows a single physical server to run multiple separate virtual machines (VMs).

- The hypervisor manages these VMs and gives each one its own CPU, memory, storage, and network resources.
- Finance, healthcare, and student applications can therefore run on different VMs on the same physical server.
- Each VM works independently, so one workload normally cannot access another workload's data.
- This helps the cloud provider share hardware safely while keeping workloads separate.

Two Major Attack Surfaces :

- Hypervisor: If the hypervisor has a security weakness, an attacker may break out of one VM and try to access other VMs.
- Virtual Network: A poorly configured virtual network may allow unwanted communication between different workloads, leading to data leakage.

Conclusion :

- Virtualization provides effective isolation, but the hypervisor and virtual network must be properly secured to protect sensitive finance, healthcare, and student data.

3. Study the following VM host utilization snapshot and recommend corrective actions: CPU 92%, memory 88%, storage I/O latency 35 ms, average VM boot time 170 s. Explain the likely bottlenecks.

VM Host Utilization Analysis :

- CPU – 92%: CPU usage is very high, which means the host is close to its processing limit. This can slow down multiple VMs.
- Memory – 88%: High memory usage may cause memory pressure and swapping, which reduces VM performance.
- Storage I/O Latency – 35 ms: The high latency shows that storage operations are taking longer than expected. This is a major storage bottleneck and can affect VM response time.
- VM Boot Time – 170 s: The long boot time is likely caused by high CPU and memory usage, along with slow storage I/O.

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Likely Bottlenecks :

- CPU: Overloaded processor.
- Memory: Insufficient available RAM.
- Storage: Slow disk I/O is the most noticeable bottleneck.

Corrective Actions :

- Move some VMs to another host to reduce CPU and memory load.
- Increase RAM and CPU capacity if the workload is continuously high.
- Use faster SSD/NVMe storage to reduce I/O latency.
- Balance workloads using VM migration or load balancing.
- Monitor resource usage regularly to prevent performance degradation.

Conclusion :

The host is overloaded, with CPU, memory, and especially storage I/O contributing to the high VM boot time.

4. Analyze how Software Defined Datacenter architecture improves agility compared with a traditional datacenter. Support your answer with compute, storage, and network virtualization considerations.

- SDDC manages the datacenter using software instead of depending mainly on physical hardware.
- Compute Virtualization: Physical servers are divided into virtual machines. VMs can be created, moved, or resized quickly according to workload needs.
- Storage Virtualization: Storage from different devices is combined into a common pool and can be allocated whenever required.
- Network Virtualization: Network resources are controlled through software, so new networks and configurations can be created quickly.
- Traditional Datacenter: It mainly uses physical servers, storage, and networking devices, so changes usually require manual configuration and more time.
- Agility: SDDC provides faster resource provisioning, easier scaling, automation, and better resource utilization.
- Management: Compute, storage, and network resources can be monitored and controlled from a centralized system.

Conclusion:

SDDC is more agile than a traditional datacenter because virtualization and software-based management allow resources to be changed, scaled, and deployed quickly with less manual work.

5. A multi-cloud federation project fails due to identity mismatches between providers. Analyze the issue and propose a secure identity and privacy design using federation concepts.

Problem :

- In multi-cloud environments, each provider may use different user IDs, roles, and authentication methods.
- This can cause identity mismatch and access problems.
- A user may have different permissions in different cloud providers.

Identity Federation :

- Use a central Identity Provider (IdP) to manage all users.
- Cloud providers act as Service Providers (SPs).
- Use SAML or OpenID Connect (OIDC) to securely exchange identity information.
- Single Sign-On (SSO) allows users to log in once and access multiple clouds.

Role Management :

- Use role mapping to match user roles across different cloud providers.
- Apply Role-Based Access Control (RBAC) to give users only the required permissions.
- Use Multi-Factor Authentication (MFA) for additional security.

Privacy Protection :

- Share only the minimum required user information with each cloud.
- Encrypt identity data during transmission and storage.
- Avoid sharing sensitive personal information unless it is necessary.
- Maintain audit logs to track user access and activities.

Example :

- A student logs in through the organization's central IdP.
- The IdP verifies the student and sends a secure identity token to different cloud providers.
- Each provider maps the user to its own Student role and permissions.
- The student can access only the required services across the clouds.

Conclusion :

- A combination of IdP, SSO, SAML/OIDC, role mapping, MFA, encryption, and minimum data sharing provides secure and privacy-aware multi-cloud federation.
- It reduces identity mismatches and makes cross-cloud access easier and safer.